VIKTOR CHALLENGE · TUM.AI · BUILD THE ROUTER

Bayern CodeWerk

–35% of the bill on a held-out chunk we had never seen — and an honest account of what our own numbers will not support

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OBJECTIVE

The trade-off we chose

Input cost per task, holding outcome quality. It is the only axis this export supports honestly: no timings ship, so latency is unmeasurable, and no quality labels ship, so quality has to be constructed — we treat it as a *constraint*, not a target.

THE SIGNAL

What the router looks at

Structure in the traces

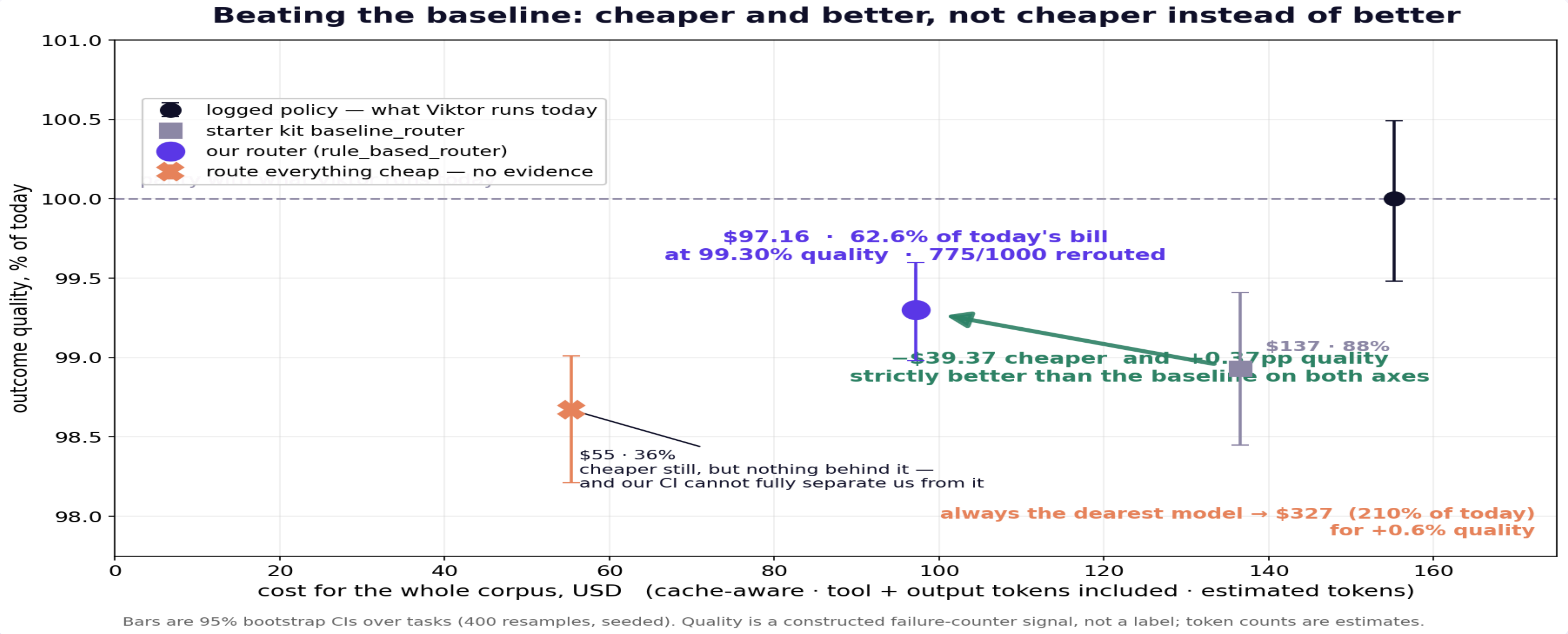
Risk signals, not prompt wording. The router reads the tool signature, whether a stakes tool was called, whether the last result was an error, and whether it is stuck in a retry loop. It starts on a price ladder and moves up or down by named steps. **Every decision reports which rules fired.**

Shown on a real trace

Viktor does not route per task today — a human picks one model per workspace. So recurring cron jobs drifted across price tiers on their own: **90 job families, 69 of them on two or more tiers.** Same job, different model. That accident is what lets us measure a model that never ran.

THE ONE CHART

Cost–quality frontier (full public dataset · n=1000 · cache-aware)



Held out: chunk 02 arrived this morning. Policy fitted on chunk 01, applied cold.
build split $155.23 → **$97.16 (62.6%)** at 99.30% · **held-out** $139.32 → **$90.29 (64.8%)** at 98.83%, 95% CI [98.41, 99.23] · 744/1000 rerouted, still all nine models.
Baseline 88% / 87%. Always-dearest 210% for +0.6% quality.

HONESTY

The off-policy estimate — and where it breaks

Method — matched same-job comparison. Recurring crons drifted across price tiers on their own, so difficulty is held fixed *by construction*, not by a similarity score. Quality is a failure-counter signal over 10,422 tool calls; 95% bootstrap CIs, 400 resamples, seeded.

Cache trap — priced cache-aware throughout; on this export each row is one complete task, so there is no shared prefix to lose. We also count what the kit omits — tool tokens and output — so **the true bill is $155, not $87**, and every percentage here is of that larger number.

Weakest point — our cost claim generalised; our quality claim did not. On the held-out chunk the starter baseline scores **0.53pp better** than us, CI [−0.90, −0.13] — it excludes zero, so that is real. And a fluent, confident, *wrong* answer with no failed tool call scores perfectly with us, which is why "route everything cheap" scores *above* us on held-out data: our signal rewards the models the operator only trusted with easy work. **We are not claiming a quality win. We are claiming a large, reproducible cost cut and an estimator we can bound.**

CLOSE

Viktor has no per-task routing today. Adding it is close to free.

Held-out: $139.32 → $90.29 · 64.8% of the bill at 98.8% of current quality

–35% on data we had never seen · 744/1000 rerouted · all nine models in play

Known weakness: on held-out data our quality estimate favours the baseline over us by half a point, and favours routing everything cheap over both. We think that is selection bias in the estimator, not truth — but we cannot prove it with this data, so we are not claiming otherwise. **Next:** judge-model rescoring on the tasks where the policies disagree.